

Climate Science, UNFCCC & Paris Agreement

GHG physics, UNFCCC–Kyoto–Paris arc, NDCs/COP and India pledges — rising #1 environment portion.

Portion focus ~88% · Topic: env-climate

PYQ themes: Paris Agreement · NDCs · COP · GHGs · Mitigation/adaptation

Why this article (PYQ probability)

Ranked among the highest next-exam-chance portions (~88%, rising). Appeared across 2016–2025; late-window COP/NDC weight exceeds early years. Parent topic env-climate ~91%.

Years appeared (synthesis): 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

Why UPSC asks this

Blends durable science with living treaty politics and India's diplomacy — ideal for mixed static–CA papers.

Why this portion is high-chance for the next exam

Ranked among the highest next-exam-chance portions (~88%, rising). Appeared across 2016–2025; late-window COP/NDC weight exceeds early years. Parent topic env-climate ~91%. Within env-climate, examiners return to climate science and the UNFCCC/Paris regime because it yields clean MCQ discriminators: definitions, matching lists, and statement–reason traps. These notes are multi-page depth material — original study synthesis for prioritisation, not NCERT/UPSC verbatim text and not an official prediction.

Use the chance % to sequence study hours, then widen to neighbouring syllabus areas. PYQ theme tags driving selection: Paris Agreement, NDCs, COP, GHGs, Mitigation/adaptation. Approximate appearance years in the 2016–2025 educational synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025.

Read this block slowly: high-chance portions repay depth because UPSC recycles the same discriminators with fresh stems. Your job is not to memorise paragraphs but to own triggers — dates, articles, site–feature pairs, formulae, treaty purposes — so elimination is automatic. After each subsection, pause and write two keywords on rough paper; if you cannot, reread only that subsection.

Core map of climate science and the UNFCCC/Paris regime

Enhanced greenhouse effect: Anthropogenic GHGs increase infrared absorption in the troposphere, raising global mean temperatures beyond the natural greenhouse baseline. Major GHGs: CO₂ (long-lived), CH₄ (higher GWP, shorter life), N₂O, and F-gases. GWP: Compares warming of a gas to CO₂ over a chosen time horizon.

Carbon sinks: Oceans, forests and soils remove CO₂; not infinite offsets. UNFCCC 1992: Framework convention creating COP process and CBDR-RC. Kyoto Protocol: Historic Annex-I style quantified targets with flexibility mechanisms.

If you can explain each point aloud in sixty seconds without notes, elimination speed in Prelims rises sharply. Watch absolute words in options (always/never/only) — often wrong where processes are multi-causal or jurisprudentially nuanced.

Draw the core map once from memory at the end of this sitting. Compare with the text. Patch only the missing nodes. That single retrieval cycle is worth more than two passive rereads of the same page.

Depth focus: Enhanced greenhouse effect

Enhanced greenhouse effect — Anthropogenic GHGs increase infrared absorption in the troposphere, raising global mean temperatures beyond the natural greenhouse baseline. In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Major GHGs — CO₂ (long-lived), CH₄ (higher GWP, shorter life), N₂O, and F-gases. Contrast the two ideas

above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: GWP

GWP — Compares warming of a gas to CO₂ over a chosen time horizon In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Carbon sinks — Oceans, forests and soils remove CO₂; not infinite offsets Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: UNFCCC 1992

UNFCCC 1992 — Framework convention creating COP process and CBDR-RC In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Kyoto Protocol — Historic Annex-I style quantified targets with flexibility mechanisms Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: Paris Agreement 2015

Paris Agreement 2015 — Universal NDCs; temperature goals well below 2C / pursue 1.5C In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Global Stocktake — Collective progress check to inform stronger NDCs Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: Transparency framework

Transparency framework — Reporting/review rules building trust In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Mitigation — Actions that reduce emissions or enhance sinks Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: Adaptation

Adaptation — Adjustments to limit harm from climate impacts In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Loss and damage — Agenda on residual irreversible harms Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: CBDR-RC

CBDR-RC — Common but differentiated responsibilities and respective capabilities In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

India narrative — Equity, intensity and non-fossil pledges, net-zero framing — verify latest NDC numbers Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: Montreal contrast

Montreal contrast — Ozone treaty — do not merge into UNFCCC In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Co-benefits — Air quality and energy security gains from climate action Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: Extremes literacy

Extremes literacy — Warming shifts probabilities of heat extremes In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Ocean acidification — CO₂ dissolution alters carbonate chemistry Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Depth focus: COP theatre

COP theatre — Revise last 2–3 COP headlines as CA overlay In exam stems, expect this idea to appear either as a direct definition or as one statement in a multi-statement cluster.

Carbon budget idea — Finite CO₂ consistent with a temperature target Contrast the two ideas above: what would a wrong option swap between them?

Micro-drill: write one correct statement and one incorrect statement using only these ideas; label why the incorrect one fails. Keep language tight — UPSC rewards precision on climate science and the UNFCCC/Paris regime more than ornamental vocabulary. If this pair links to a map or article number, add that trigger in the margin of your PDF printout.

Comparison tables — memorise differences, not essays

Tables win Prelims because UPSC loves correctly matched pairs and multi-statement sets. Study the table(s) embedded with this article until you can reproduce them on rough paper in under four minutes. Then invent two false rows and practise spotting them — that simulates assertion traps.

After table mastery, return to narrative sections only for rows you keep missing. Depth here is intentional: thin skim notes lose marks on discriminators. When a table row has a date, article, or place, that cell is a trigger — circle it.

Must-know India / applied angle

India emphasises equity and development space while expanding renewables and long-term net-zero narratives. Keep three pledge anchors updated from official NDC documents. Keep a rolling one-page CA addendum dated by month; do not rewrite your static spine every week. Mains crossover exists, but Prelims primarily rewards precise factual edges and definitions.

Link each applied example back to a static hook so current affairs does not float free of the syllabus. If a news item cannot attach to a hook within ten seconds, park it in a low-priority list. Rehearse a 45-second spoken India angle the day before any mock that might touch this portion.

How UPSC typically frames questions (PYQ angle)

Treaty design statements; GHG/sinks science; COP finance language; India pledges. Tactic: eliminate anachronisms and category errors first (wrong institution, wrong treaty family, wrong century, wrong formula base). Then test remaining options against table rows. Finally, beware extra absolute adjectives.

Assertion–reason items often smuggle a true statement with a false causal link — judge the link, not only the facts. Matching lists reward uniqueness: start with the most distinctive pair. Themes: Paris Agreement, NDCs, COP, GHGs, Mitigation/adaptation. Years signal in this hub's synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025.

Common traps and how to dodge them

Trap: Treating Kyoto and Paris as identical. Trap: Confusing Montreal with UNFCCC. Trap: NDCs as mere press releases.

Trap: Net zero as overnight zero emissions. Trap: Collapsing mitigation/adaptation/L&D. Trap: Infinite forest offsets.

Build a personal trap diary: after each mock, add one line under this heading. Reread only the trap diary the night before the paper for this portion. Teach one trap to a peer in one minute — teaching exposes fuzzy understanding.

Worked revision method (make the PDF earn its pages)

Day 1: core map + table; close notes; reproduce table. Day 2: India angle + traps; 15 MCQs. Day 3: PYQ-style self-questions from headings only. Day 4: teach the topic aloud for eight minutes. Day 5: mixed quiz with neighbouring topics to prevent silo memory.

Print the PDF or use in-app Download PDF. Annotate margins only with triggers you forget — not entire yellow pages. Spaced retrieval beats rereading: hide the PDF and write bullets from memory, then check. In the final 15 days, this portion should be in 'table + traps + must' mode, not first-read mode.

Quick synthesis for the exam hall

GHG science ! UNFCCC/Kyoto/Paris ! NDC/stocktake ! India anchors ! Montreal trap. Portion focus score used in-app: ~88%. If time is scarce in the last week, reread Must remember + tables + traps only; return to long sections for weak rows.

Carry a one-line spine on rough paper in the first minute of the paper if this topic appears early — then eliminate. Do not invent facts under pressure; prefer elimination from your table.

Extended notes — depth for multi-page mastery

Enhanced greenhouse effect. Anthropogenic GHGs increase infrared absorption in the troposphere, raising global mean temperatures beyond the natural greenhouse baseline Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Major GHGs. CO₂ (long-lived), CH₄ (higher GWP, shorter life), N₂O, and F-gases Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. GWP. Compares warming of a gas to CO₂ over a chosen time horizon Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Carbon sinks. Oceans, forests and soils remove CO₂; not infinite offsets Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. UNFCCC 1992. Framework convention creating COP process and CBDR-RC Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin.

Kyoto Protocol. Historic Annex-I style quantified targets with flexibility mechanisms Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Paris Agreement 2015. Universal NDCs; temperature goals well below 2C / pursue 1.5C Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Global Stocktake. Collective progress check to inform stronger NDCs Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Transparency framework. Reporting/review rules building trust Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Mitigation. Actions that reduce emissions or enhance sinks Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin.

Adaptation. Adjustments to limit harm from climate impacts Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Loss and damage. Agenda on residual irreversible harms Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. CBDR-RC. Common but differentiated responsibilities and respective capabilities Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. India narrative. Equity, intensity and non-fossil pledges, net-zero framing — verify latest NDC numbers Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Montreal contrast. Ozone treaty — do not merge into UNFCCC Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin.

Co-benefits. Air quality and energy security gains from climate action Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Extremes literacy. Warming shifts probabilities of heat extremes Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Ocean acidification. CO₂ dissolution alters carbonate chemistry Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. COP theatre. Revise last 2–3 COP headlines as CA overlay Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin. Carbon budget idea. Finite CO₂ consistent with a temperature target Pair this with one contrasting point so options cannot blur categories. Add a keyword trigger in your margin.

This extended module exists so downloads are clearly longer richer than skim notes. Read it in two sittings if needed; do not skim on day one and never return. After finishing, close the article and write a 12-line summary from memory; compare; patch gaps only. Then solve at least ten mixed MCQs tagging each error to a heading in this PDF.

Error log template (copy into your notebook)

For every missed question related to this portion, record: (1) stem gist, (2) wrong choice, (3) correct choice, (4) misconception in five words, (5) table row or must-remember line that would have saved you. Review the error log before the next mock. If the same misconception repeats thrice, rewrite that depth-focus section in your own words on a fresh page.

This metacognitive loop is how multi-page notes convert into marks rather than shelf weight.

Self-check questions (close the book first)

1) Define climate science and the UNFCCC/Paris regime in two sentences without buzzword salad. 2) Recite the comparison table rows from memory. 3) List three traps and the correct mental fix for each. 4) Give two India examples and one international or cross-topic anchor. 5) Write one incorrect statement UPSC might plant — then correct it. 6) Time yourself: explain the exam spine in 90 seconds. 7) List five triggers (dates/articles/sites/formulae) without peeking.

Pass criteria: 6/7 without peeking. Then move to mixed MCQs. Failing is useful if you patch immediately — that is the point of depth notes.

Treaty design contrast

Feature	Kyoto (classic)	Paris
Participation	Annex-I focus historically	Near-universal NDCs
Target style	Top-down quantified	Bottom-up NDCs
Temperature goal	Not Paris-style 1.5/2 framing	Well below 2C; pursue 1.5C
Ambition cycle	Commitment periods	NDC cycles + Stocktake

Vocabulary triad

Term	Focus	Prelims cue
Mitigation	Cut/sequester GHGs	Renewables, sinks
Adaptation	Live with impacts	EWS, resilient infra
Loss & damage	Residual harm finance/politics	Vulnerable nations

Map / list facts

- Himalayan cryosphere
- Coastal vulnerability
- SIDS L&D politics
- India renewables geography

Common traps

- ' Treating Kyoto and Paris as identical
- ' Confusing Montreal with UNFCCC
- ' NDCs as mere press releases
- ' Net zero as overnight zero emissions
- ' Collapsing mitigation/adaptation/L&D
- ' Infinite forest offsets

Must remember

- UNFCCC!'Kyoto!'Paris differences
- GHGs + GWP idea

- Paris NDCs/stocktake
- CBDR-RC
- Mitigation vs adaptation vs L&D
- Refresh India NDC numbers
- Montreal¹ UNFCCC

Quick revision checklist

- & Redraw Kyoto vs Paris table
- & List GHG + sink
- & Define mitigation/adaptation/L&D
- & Update India pledges
- & Last 2 COP bullets

Educational prioritisation heuristic — not an official UPSC prediction. © UPSC IAS Study Hub original notes.

